

NADIA LOTFI

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EDUCATION

Northwestern University

Evanston, IL

Bachelor of Science: Biomedical Engineering

2024 – 2028

- Cumulative GPA: 3.78/4.0
- Relevant Coursework: Biomedical Signals & Circuits, Bioelectronics, Experimental Design, Mechanical Design & Manufacturing, Biomedical Systems Analysis, Materials Science, Thermodynamics, Molecular & Cell Biology, Python Programming

RESEARCH EXPERIENCE

Bin Jiang Lab

Chicago, IL

Research Intern

April 2025 – January 2026

McCormick Summer Research Grant (\$4,500)

- Designed, optimized, and performed 10+ experiments investigating mitochondrial transplantation as a therapeutic approach for diabetic endothelial dysfunction, assessing viability, mitochondrial function, inflammatory markers, and migration
- Applied in vivo diabetic wound models and histology to evaluate aims, analyzed 500+ microscopy images with ImageJ and Python
- Collaborate with team to prepare technical documentation, data, figures, and manuscript revisions for an in-progress publication

Magnetic Resonance Technology Discovery Lab

Chicago, IL

Research Intern

April 2023 – September 2023

- Performed T1 relaxometry analysis on gadolinium contrast phantoms using the Hyperfine Swoop low-field MRI through precision testing, data collection, and analysis with MATLAB and qMRLab for signal fitting and relaxation curve analysis
- Evaluated low-field MRI systems as accessible diagnostic tools for under-resourced healthcare settings

WORK EXPERIENCE

Segal Design Institute

Evanston, IL

Shop Trainer

September 2025 – Present

- Support student innovation by advising on design strategies, CAD, and material selection for engineering projects
- Instruct engineering students on the safe and effective use of machining tools, including laser cutters, drill presses, mills, lathes, bandsaws, and hand tools, while providing technical guidance on machine selection and design-for-manufacturing considerations

Northwestern Academic Support and Learning Advancement

Evanston, IL

Peer Tutor

September 2025 – Present

- Provide one-on-one and group tutoring for core engineering courses including differential equations, linear algebra, and physics
- Develop problem-solving strategies for students by breaking down complex mathematical concepts, guiding exam preparation, and tailoring explanations to diverse learning styles, helping students build long-term study and analytical skills

PROJECTS

Oscillometric Blood Pressure Monitoring System

Evanston, IL

Project Member

April 2026 – June 2026

- Designed an oscillometric blood pressure measurement system incorporating pressure sensing, instrumentation amplification, and analog filtering circuitry to isolate arterial pulse signals, applying time-domain and frequency-domain signal analysis
- Processed physiological signals in Python to estimate heart rate, arterial pressure, and blood pressure from cuff pressure data

Bicycle Caliper Brake Design

Evanston, IL

Project Member

April 2026 – June 2026

- Designed, manufactured, and tested a lightweight bicycle caliper brake system to satisfy ISO braking and structural requirements
- Performed finite element analysis (FEA), topology optimization, and analytical stress and deflection calculations to improve stiffness, reduce stress concentrations, and minimize material usage, achieving compliant performance while maintaining a lightweight structure

Once Upon a Tile

Evanston, IL

Project Member

April 2025 – June 2025

- Designed an interactive robotics system in collaboration with Misericordia Heart of Mercy through iterative design and testing to enhance creative learning for adults with cognitive and developmental disabilities, focusing on accessibility and intuitiveness
- Programmed a robot to interpret and follow user-drawn trajectories across modular whiteboard puzzle tiles

Legacy PopUp Display

Evanston, IL

Project Member

January 2025 – March 2025

- Designed a portable, modular display system for the Shorefront Legacy Center to more effectively share Black community histories
- Engineered protective modules capable of securely storing and presenting both 2D and 3D media, ensuring durable exhibition

ADDITIONAL INFORMATION

Skills: Computational and Design Software (Python, MATLAB, SolidWorks, ImageJ, qMRLab, Prism, Microsoft Office), Topology Optimization, Finite Element Analysis, Prototyping and Fabrication Tools, Statistical Analysis, Cell Culture, Histology and Staining

Activities: Persian Organization (President), Society of Women Engineers (Outreach), Biomedical Engineering Society (Internal Comms)

Awards: McCormick Summer Research Grant (\$4,500), National Merit Scholarship (\$4000), Northwestern Dean's List High Honors